



Introduction to Brain and Cognition

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Course Content

- Learning Outcomes:
 - Broad knowledge and Appreciation of Brain and Cognition
 - **Book:** Daniel Reisberg (2019). *Cognition: exploring the science of the mind*. 7th Edition. W. W. Norton & Company, NY, USA
 - **Book:** Woo-Kyoung Ahn (2022). *Thinking 101: How to Reason Better to Live Better*. Macmillan, NY, USA
 - Brief acquaintance with the Research interests of CogSciLab (CSL) faculty!

Learning Assessment

- Assignments: 40% [Best 4 of 5/6]
- Quiz: 20%
- Final: 40%

READING ASSIGNMENT: Chapter 1, Reisberg (2019). Cognition

Chapter 1: The Science of the Mind

- The Scope of Cognitive Psychology
 - Cognitive Psychology when first launched was about *Scientific study of knowledge*.
 - How is knowledge acquired? How is knowledge retained so that it's available when needed? How is knowledge used — whether as a basis for making decisions or as a means of solving problems?
 - But more broadly, Cognitive psychology is concerned with how people remember, pay attention, and think.
 - The importance of all these issues arises partly from the fact that most of what we do, say, and feel is guided by things we already know (*Memory*).

Chapter 1: The Science of the Mind

- The Scope of Cognitive Psychology
 - One example is our comprehension of a simple story, which turns out to be heavily influenced by the knowledge we supply.
 - *Lakshmi wanted to bring Nitin a present. She shook her piggy bank. It made no sound. She went to look for her mother.*



Chapter 1: The Science of the Mind

- The Cognitive Revolution
 - Cognitive psychology emerged as a separate discipline in the late 1950s, and its powerful impact on the wider field of psychology has led many academics to speak of this emergence as the *cognitive revolution*.

Chapter 1: The Science of the Mind

- The Cognitive Revolution
 - *Introspection*:
 - One predecessor of cognitive psychology was the 19th-century movement that emphasized *introspection* as the main research tool for psychology.
 - But psychologists soon became disenchanted with this movement for several reasons:
 - Introspection cannot inform us about unconscious mental events;
 - Even with conscious events, claims rooted in introspection are often untestable because there is no way for an independent observer to check the accuracy or completeness of an introspective report.

Chapter 1: The Science of the Mind

- The Cognitive Revolution
 - *Behaviorism:*
 - The behaviorist movement rejected introspection as a method, insisting instead that psychology speak only of mechanisms and processes that are objective and out in the open for all to observe.
 - However, evidence suggests that our thinking, behavior, and feelings are often shaped by our perception or understanding of the events we experience.

Chapter 1: The Science of the Mind

- The Cognitive Revolution
 - *Behaviorism*:
 - This is problematic for the behaviorists: *Perception* and *understanding* are exactly the sorts of mental processes that the behaviorists regarded as subjective and not open to scientific study.
 - *Pass the salt* example for impossibility of a complete behaviorist psychology

Chapter 1: The Science of the Mind

- The Cognitive Revolution
 - *Cognitivism:*
 - To study *mental events*, psychologists have turned to a method in which one focuses on observable events but then asks what (invisible) events must have taken place in order to make these (visible) effects possible.

Chapter 1: The Science of the Mind

- Research in Cognitive Psychology
 - *Many factors contributed to the emergence of cognitive psychology in the 1950s and 1960s.*
 - **Miller's** limits on working memory and chunking.
 - **Tolman's** research demonstrated that even in rats, learning involved the acquisition of new knowledge and not just a change in behavior.
 - **Chomsky** argued powerfully that a behaviorist analysis was inadequate as an explanation for language learning and language use.

Chapter 1: The Science of the Mind

- Research in Cognitive Psychology
 - *Many factors contributed to the emergence of cognitive psychology in the 1950s and 1960s.*
 - **Gestalt psychologists** emphasized the role of the perceiver in organizing his or her experience.
 - **Bartlett's** research showed that people spontaneously fit their experiences into a mental framework, or schema.

Chapter 1: The Science of the Mind

- Research in Cognitive Psychology
 - Early theorizing in cognitive psychology often borrowed ideas from computer science, including early work on artificial intelligence.
 - Cognitive psychologists rely on a diverse set of methods and collect many types of data.
 - Measures of the quality of someone's performance (**Accuracy**),
 - Measures of response speed (Reaction Time **RT**),
 - Methods that allow us to probe the underlying biology (Neuroimaging).